

Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25EVT122 - MODELING AND ANALYSIS OF ELECTRICAL MACHINES

(For EVT)

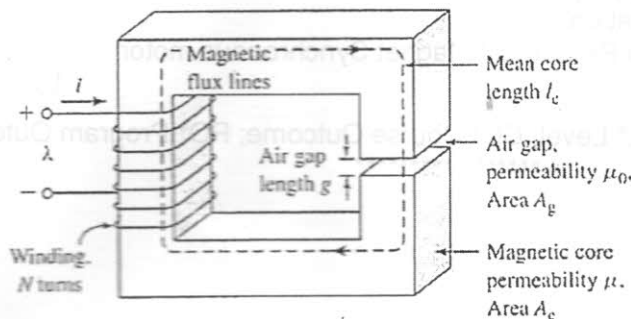
Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit-I** and **Unit-II** and **One full** question from **Unit-III**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Describe space phasors. Elaborate the production of a travelling mmf wave by combination of three standing mmf waves in the stator of an ac three-phase synchronous motor. | 6 | L3 | 1 | 1,4 |
| b) In the magnetic circuit shown below, cross-sectional areas of both the core and air-gap are 12 cm^2 . The air-gap length is 0.04 cm while the mean length of flux path in the core is 40 cm . The winding includes 600 number of turns. Taking the relative permeability of core steel as 60,000, find reluctance of core iron and air-gap. Considering core flux density as 1.2 T , find the core flux and magnetizing current. | | | | |



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|---|----|----|---|-----|
| c) Why linearization techniques for small perturbations are necessary? Apply this technique to develop a small signal model for DC series motor. | 7 | L3 | 1 | 1,4 |
| 2. a) Derive the mathematical model of separately excited DC motor and hence obtain the Transfer Function. | 10 | L3 | 2 | 1,4 |
| b) Derive the power balance expression in terms of electrical parameters for a singly-excited primitive motor with salient structure. | 10 | L3 | 1 | 1,4 |
| 3. a) Derive the Kron's primitive machine-voltage, current equations and hence predict the mechanical power developed by the machine. | 12 | L3 | 1 | 1,4 |
| b) A separately excited dc motor is delivering rated torque at rated speed. Find the efficiency of the motor at this operating point. The details of the machine are as follows:
1500kW, 600V, rated current=2650A, 600rpm, Brush voltage drop=2V, Field power input=50kW, $R_a=0.003645\Omega$, $L_a=0.1\text{mH}$, Machine frictional torque coefficient=15 N-m/(rad/sec). Field current is constant and the armature voltage is variable. | 8 | L3 | 2 | 1,4 |

Unit – II

- | | | | | |
|---|----|----|---|-----|
| 4. a) (i) Show that Park matrix is power invariant
(ii) Using a power invariant transformation with θ for the transformation given by $\theta = \omega_s t + \delta$, determine f_{abc} when | | | | |
| $f_{abc} = \left[f_m \sin(\omega_s t + \theta), f_m \sin\left(\omega_s t + \theta - \frac{2\pi}{3}\right), f_m \sin\left(\omega_s t + \theta + \frac{2\pi}{3}\right) \right]$ | 10 | L4 | 3 | 1,4 |



b)	With the help of diagrams, analyse the following	10	L3	4	1,4
	(i) Regulating transformer for voltage control (ii) Regulating transformer for phase angle control				
5. a)	Derive an expression for the electromagnetic torque of an induction motor using dq transformation theory.	10	L3	3	1,4
	b) A three-phase star-connected balanced voltage source is supplying a three-phase symmetrical load. The load on each phase comprises of an identical series combination of a resistor and an inductor. Obtain the state equations in 0dq frame of reference using power invariant Park transformation.				
6. a)	Derive the π model of a transformer operating with off nominal turns ratio.	10	L3	4	1,4
	b) Derive per unit model of an induction motor using dq modeling theory.				

Unit – III

7. a)	Derive the expressions for stator self inductances and stator mutual inductances for 3phase salient pole synchronous motor.	10	L3	5	1,4
	b) Derive the expression* for electromagnetic Torque developed in synchronous machine in rotor reference frame.				
8. a)	Derive the equivalent circuit of 3phase synchronous motor in rotor reference frame using Park equations.	10	L3	5	1,4
	b) Develop the dynamic model of a Permanent Magnet Synchronous motor in rotor reference frame.				

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
